RAG Assignment: Baseline vs. Re-Ranking

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Code: https://github.com/Sourav-Nath-01/RAG-Assignment-Baseline-vs.-Re-Ranking

Setup

Both pipelines search the same 565 GDPR chunks with one shared prompt, so Answer 1 and Answer 2 differ only in which three chunks the LLM sees.

- **Document:** GDPR (Regulation (EU) 2016/679), official 88-page EUR-Lex PDF. Its 173 non-binding recitals paraphrase 99 binding articles full of "unless…" exceptions and cross-references, a rich source of distractors.
- **Chunks:** at most 150 words, 30-word overlap, never crossing an article: 565 chunks, all under MiniLM's 256-token limit.
- **Models:** bi-encoder all-MiniLM-L6-v2 with a FAISS IndexFlatIP index (cosine); cross-encoder ms-marco-MiniLM-L-6-v2; Gemini 2.5 Flash at temperature 0, told to answer only from the excerpts or reply "Not found in the provided context."

1. Architecture comparison

A bi-encoder turns the query and each chunk into vectors **separately** and compares them afterwards; a cross-encoder reads the query and the chunk **together** and outputs one relevance score.

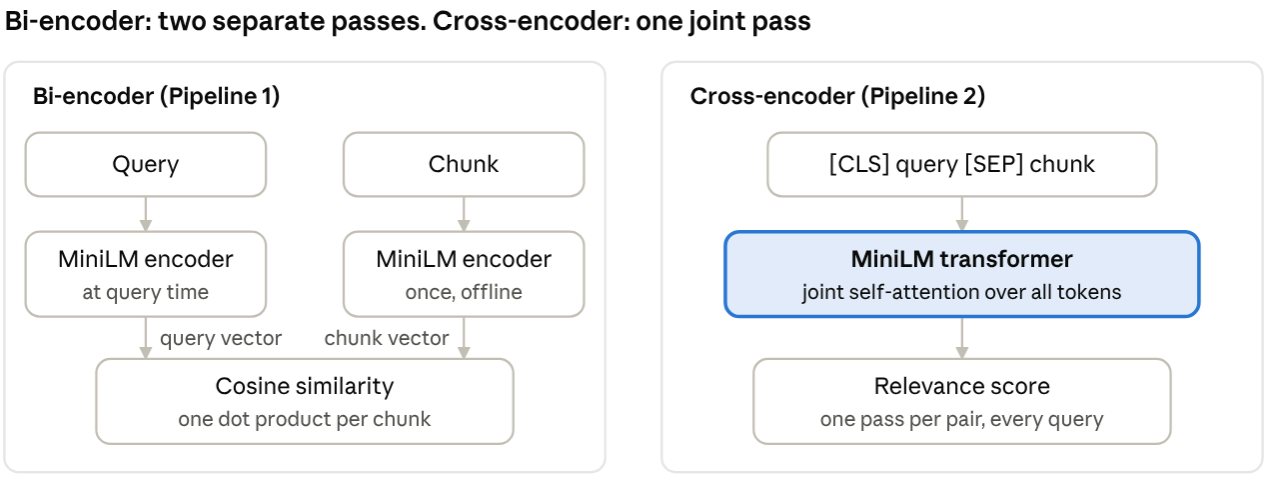


Figure 1. How each model scores a (query, chunk) pair.

Bi-encoder: each chunk is encoded once, offline, into a 384-d vector; at query time only the query is encoded and compared by cosine. This is fast, but a chunk's meaning is squeezed into one vector before the query is known, so shared topic words dominate and words like "not" or "unless" barely move it.

Cross-encoder: `[CLS] query [SEP] chunk` is one input, so self-attention links every query token to every chunk token and the model can judge whether the chunk actually answers the question. Nothing can be precomputed, though: every query needs one full forward pass per candidate. Hence the combination: the bi-encoder for cheap recall (10 candidates), the cross-encoder for precise ranking (the best 3).

2. Qualitative and rank shift analysis

In all four failure queries the answer chunk sat at bi-encoder rank 4–5, just outside the Top 3; the cross-encoder moved it to rank 1–2 and Answer 2 became correct. I found the cases among 109 candidate questions, and each gold chunk is fixed by an exact phrase of the regulation, never by looking at the rankings.

ID	Failure type	Gold (chunk)	Rank shift	Answer 1	Answer 2
KT-1	Keyword trap	Art. 79(1) (528)	#4 → #2	Misstated rule	Correct
EX-1	Exception	Art. 33(1) (366)	#5 → #2	Abstained	Correct
NEG-1	Negation	Art. 2(2) (259)	#5 → #1	Abstained	Correct
DOC-1	Recitals beat article	Art. 12(2) (301)	#4 → #1	Omitted clause	Correct
LIM-1	Re-ranking hurts	Art. 14(5) (317)	#4 → #9	Correct	Abstained
LIM-2	Recall limit	Art. 83(4) (538)	#21 → –	Abstained	Abstained

Table 1. Gold-chunk rank shift and answer verdicts for each failure query.

- **KT-1, keyword trap.** *"Is a person allowed to complain to a court instead of the regulator?"* "Court" and "regulator" pull in Recital 143 and Art. 78 (suing the regulator after a rejected complaint); Art. 79(1) answers in other words ("an effective judicial remedy … without prejudice to … the right to lodge a complaint"). Answer 1 misstated the rule as "court only after a rejected complaint"; Answer 2 said one can sue the controller directly.
- **EX-1, exception.** *"Does every data breach have to be reported to the supervisory authority?"* The baseline got breach chunks but not the "unless … unlikely to result in a risk" carve-out in Art. 33(1); Answer 2 gave the 72-hour rule and the exception.
- **NEG-1, negation.** *"Which activities fall outside the scope of the regulation?"* The bi-encoder matched generic words in Art. 57, Recital 73 and Art. 40; the cross-encoder linked "outside the scope" to "does not apply to" in Art. 2(2).

- **DOC-1, recitals outrank the article.** *"Is a controller allowed to refuse to act on a request when it cannot identify the data subject?"* Recitals 57 and 47 and Art. 11 beat Art. 12(2) ("shall not refuse … unless the controller demonstrates…"); Answer 1 omitted that the controller must prove it cannot identify the person.

Answer 1 vs. Answer 2. No answer hallucinated: with outside knowledge forbidden, a missing chunk showed up as "Not found" (EX-1, NEG-1), an omitted clause (DOC-1) or a rule from the wrong article (KT-1).

Limitations. In LIM-1 the cross-encoder preferred recitals about research in general and dropped both exemption chunks (Recital 62 #1 → #7, Art. 14(5) #4 → #9), so Answer 2 abstained. In LIM-2, Art. 83(4) names the fine only as "Articles 8, 11, 25 to 39", so it ranked #21 and was never re-ranked.

Query type	Queries	Recall@10	Hit@3, P1 → P2	MRR@10, P1 → P2
Keyword trap	67	0.63	0.46 → 0.49	0.38 → 0.42
Exception	23	0.57	0.52 → 0.52	0.37 → 0.48
Negation	19	0.84	0.68 → 0.74	0.59 → 0.68
All	**109**	**0.65**	**0.51 → 0.54**	**0.41 → 0.48**

Table 2. Retrieval quality over all 109 candidate questions (P1 = baseline, P2 = re-ranked).

The gain is modest: 7 gold chunks were promoted into the Top 3 and 4 pushed out. MRR rises more than Hit@3 because most gains happen inside the Top 3 (gold at #1: 31 → 42). For 35% of questions the answer was not in the Top 10, which no re-ranker can fix.

3. Latency profiling

Re-ranking makes retrieval about 32× slower (17.5 → 565.9 ms median), adding 0.55 s (44%) end to end; the LLM call is still the largest cost.

Stage (median, ms)	Pipeline 1 – Baseline	Pipeline 2 – Re-ranking
Bi-encoder retrieval (embed + search)	17.5	17.5
Cross-encoder, 10 pairs + sort	–	548.5
Gemini answer generation	1,226.6	1,384.1
End to end	**1,244.1**	**1,950.0**

Table 3. Median latency per query, CPU only.

Measured on an AMD Ryzen 5 3550H over 140 retrieval runs (7 queries × 20); Gemini times are real API round trips, so the 160 ms LLM gap is network noise. Re-ranking runs 10 full transformer passes (about 55 ms each) versus one short query encoding. One-off costs (model loading 16.6 s, indexing 23.8 s) are not paid per query.

4. Scaling question

A cross-encoder needs one full pass per chunk for every query, so its cost grows linearly with the collection and nothing can be precomputed. Here, scoring all 565 chunks already takes 27.9 s per query, against 0.56 s for retrieve-then-re-rank.

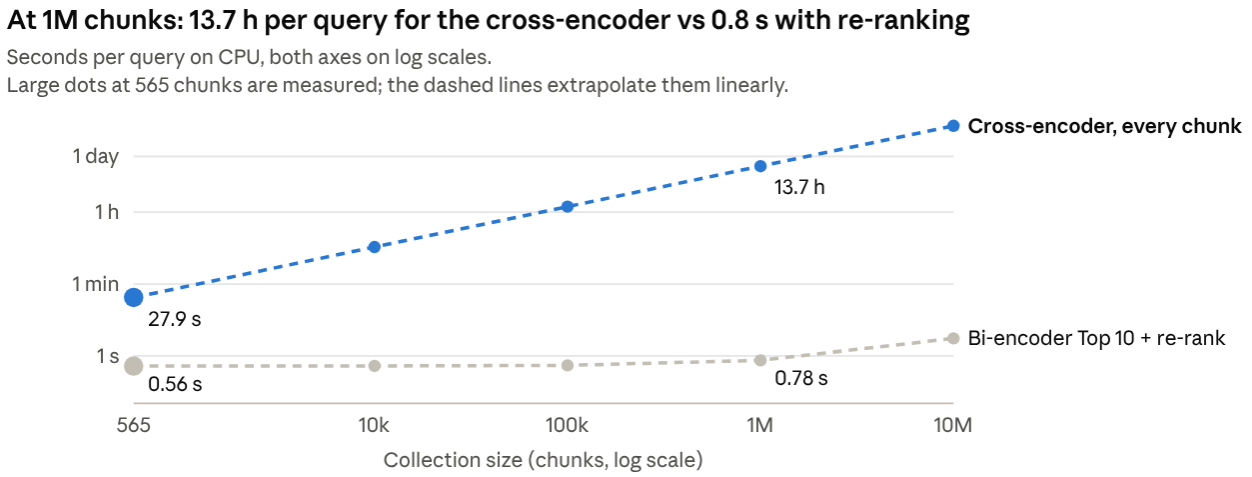


Figure 2. Seconds per query against collection size (results/scaling_extrapolation.csv).

- **No precomputation:** the score depends on the query, whereas the bi-encoder's chunk vectors are built once (23.8 s here) and reused; with an approximate index such as HNSW the search even grows sub-linearly.
- **Little extra quality:** run over all 565 chunks, the cross-encoder ranked the gold chunk of KT-1, EX-1, NEG-1 and DOC-1 at #1–#3 (no better than re-ranking the Top 10) and still missed LIM-2 (#48) and LIM-1 (#29). Re-ranking a short list keeps most of the benefit at about 2% of the cost; spare compute is better spent on recall (more candidates, hybrid keyword + dense search).